



Short Communication

Molecular epidemiology of multidrug-resistant *Acinetobacter baumannii* isolates from a hospital in NepalMasafumi Sakuma^a, Mari Tohya^b, Tomomi Hishinuma^a, Jeevan B. Sherchand^c, Teruo Kirikae^d, Tatsuya Tada^{a,*}^a Department of Microbiology, Juntendo University School of Medicine, Bunkyo-ku, Tokyo, Japan^b National Institute of Health Sciences, Kawasaki, Kanagawa, Japan^c Tribhuvan University, Kathmandu, Nepal^d Juntendo University, Juntendo Advanced Research Institute of Health Science, Bunkyo, Japan

ARTICLE INFO

Article history:

Received 4 June 2024

Revised 10 July 2024

Accepted 29 July 2024

Available online 6 August 2024

Editor: Stefania Stefani

Keywords:

Multidrug-resistant *Acinetobacter baumannii*

Carbapenemase

16S rRNA methyltransferase

Molecular epidemiology

ABSTRACT

Objectives: The emergence of multidrug-resistant (MDR) *Acinetobacter baumannii* has become a serious worldwide medical problem. This study was designed to clarify the genetic and epidemiological properties of MDR *A. baumannii* clinical isolates.

Methods: A total of 66 MDR *A. baumannii* isolates were obtained from 66 inpatients between May 2019 and February 2020 in a university hospital in Nepal. Whole genomes of these isolates were sequenced using next-generation sequencing. Phylogenetic trees were constructed from single nucleotide polymorphism concatemers. Multilocus sequence typing (MLST) and clonal complex (CC) analysis were conducted, and drug-resistance genes were identified.

Results: Of the 66 isolates, 26 harboured a gene encoding NDM-type metallo- β -lactamase, and 55 harboured a gene encoding the 16S rRNA methyltransferase, ArmA. All isolates had point mutations in the quinolone-resistance-determining regions of *gyrA* and *parC*. Phylogenetic analysis showed that 55 isolates harboured *armA*, 26 harboured *bla*_{NDM-1}, and 14 harboured *bla*_{PER-7}. Multilocus sequence typing and CC analysis revealed that 34 isolates belonged to CC2 (ST2), 10 to CC1 (nine ST1 and one ST623), and eight to CC149 (ST149). Compared to our previous study on MDR *A. baumannii* in Nepal in 2012, the isolation rate of CC2 increased, whereas that of CC149 decreased between 2012 and 2020.

Conclusions: This study indicates that MDR *A. baumannii* producing carbapenemase and 16S rRNA methyltransferase, with high resistance to carbapenems and/or aminoglycosides, are spreading in medical settings in Nepal. The genetic backgrounds of MDR *A. baumannii* isolates have shifted to international clone 2 over several years.

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1. Introduction

The emergence of multidrug-resistant (MDR) *Acinetobacter baumannii* has become a serious worldwide medical problem [1]. MDR *A. baumannii* is frequently involved in nosocomial outbreaks, particularly in intensive care units [2]. Nosocomial pneumonia caused by MDR *A. baumannii* may affect critically ill patients, making treatment more challenging [1,2]. Most MDR *A. baumannii* isolates have developed extensive resistance to

antimicrobial agents through enzymatic modification of antibiotics and mutations in target genes associated with antibiotics [3]. For example, the predominant mechanism of resistance to β -lactams in *A. baumannii* is enzymatic hydrolysis by β -lactamases, whereas resistance to aminoglycosides results from the production of aminoglycoside-modifying enzymes and 16S rRNA methyltransferases [3]. Fluoroquinolone resistance is due to mutations in the target enzymes DNA gyrase and DNA topoisomerase IV, encoded by the genes *gyrA* and *parC*, respectively [4].

Carbapenem resistance rates in *A. baumannii* strains have been rising dramatically worldwide [5]. The most common mechanism of carbapenem resistance in *A. baumannii* isolates from human infections is the acquisition of carbapenem-hydrolysing

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class D β -lactamases, including OXA-23, OXA-51-like (intrinsic carbapenem-hydrolysing class D β -lactamases), OXA-58, OXA24/40 and OXA-143 [6]. Additionally, some highly carbapenem-resistant *A. baumannii* isolates produce class B metallo- β -lactamases (MBL), including IMP, NDM, SIM, and VIM types [1]. Aminoglycoside resistance in *A. baumannii* involves the production of aminoglycoside-modifying enzymes, which are further categorized as acetyl-, adenyl-, and phosphor-transferases, based on the modification site of aminoglycosides (primarily *N*-acetylation, *O*-nucleotidylation, and *O*-phosphorylation, respectively) [7]. Another resistance mechanism involves the production of 16S rRNA methyltransferases, which confer high-level resistance to all clinically useful aminoglycosides [3]. These methyltransferases include *armA*, *rmtA*, *rmtB*, *rmtC*, and *rmtD*, with the *armA* gene also found in other Gram-negative organisms [3]. This study analysed the molecular epidemiology of MDR *A. baumannii* isolates obtained from hospitals in Nepal.

2. Materials and methods

2.1. Bacterial strains and drug susceptibility testing

A total of 66 MDR *A. baumannii* isolates, defined as strains showing resistance to carbapenem (MIC ≥ 8 μ g/mL), amikacin (MIC ≥ 64 μ g/mL), and ciprofloxacin (MIC ≥ 4 μ g/mL), were obtained from individual patients treated at a university hospital in Nepal, from May 2019 to February 2020. *A. baumannii* isolates were identified by biochemical methods and by sequencing their 16S rRNA, *gyrB*, and *bla*_{OXA-51-like} genes. Antimicrobial susceptibility testing was performed using the micro-dilution method according to the guidelines of the Clinical and Laboratory Standards Institute [8]. The breakpoint for tigecycline resistance was provided for *Enterobacteriaceae* but not for *Acinetobacter* spp. by EUCAST and the US Food and Drug Administration, although it was proposed as ≥ 4 μ g/mL [9].

2.2. Whole genome sequencing

The whole genome of each isolate was extracted using a DNeasy Blood & Tissue Kit (Qiagen, Tokyo, Japan) and sequenced using a next-generation sequencer (MiSeq; Illumina Inc., San Diego, CA). Raw reads of each isolate were assembled using CLC Genomic Workbench version 11.0.1. The whole genomes of 11 strains harbouring *bla*_{NDM-1} and/or *armA* belonging to different sequence types (ST) were extracted using Qiagen Genomic-tip 100/G and Genomic DNA Buffer Set (Qiagen) and sequenced using MinION (Oxford Nanopore Technologies, Oxford, UK). The raw reads from MiSeq and MinION were assembled using Unicycler v.0.4.7 [10].

2.3. Drug resistance and virulence genes

Drug-resistance genes were identified using ResFinder 4.5.0 (<http://genepi.food.dtu.dk/resfinder>). Virulence genes associated with biofilm formation, including *cusE*, *pgaB*, *bfmS*, *leuO*, *ompA*, and *epsA/ptk*, were analysed using CLC Genomic Workbench version 11.0.1.

2.4. Multilocus sequence typing (MLST) and phylogenetic analysis

MLST was conducted as described by protocols of Institute Pasteur MLST (<http://www.pasteur.fr/recherche/genopole/PF8/mlst/Abbaumannii.html>). Clonal complex (CC) was determined using eBURST version 3 (<http://eburst.mlst.net>). Genomic environments surrounding *bla*_{NDM-1} and *armA*, which confer high levels of resistance to β -lactams and aminoglycosides, respectively, were compared using in silico molecular cloning (In Silico Biology, Inc

Yokohama, Japan). Single nucleotide polymorphisms (SNP) of the genome sequences of all MDR isolates tested were identified by comparison with the sequence of *A. baumannii* type strain ATCC19606 (GenBank accession no. CP045110), and phylogenetic trees were constructed from the SNP concatemers using kSNP4.0.

2.5. GenBank accession numbers

The raw reads of all 66 isolates were deposited at GenBank with accession numbers DRA 018,628. The sequence data of chromosomes and plasmids in 11 *A. baumannii* isolates harbouring genes encoding NDM-1 and/or *ArmA* were deposited at GenBank with accession numbers AP031576 and AP031588 (ST1 chromosomes), AP031577, AP031578 and AP031580 (ST2 chromosomes), AP031582 (ST10 chromosome), AP031579 (ST15 chromosome), AP031586 (ST25 chromosome), AP031587 (ST149 plasmid), AP031583 and AP031585 (ST149 chromosomes), AP031584 (ST149 plasmids), and AP031581 (ST164 chromosome).

3. Results

3.1. Drug susceptibility profiles, drug resistance, and virulence genes

Of the 66 MDR *A. baumannii* isolates, 28 were from respiratory tracts, 16 from pus, nine from blood, seven from wound swabs, three from urine, and three from body fluid. All isolates were resistant to imipenem, meropenem, ceftazidime, cefotaxime, amikacin, and ciprofloxacin, but susceptible to tigecycline (Table S1). Most of the isolates were resistant to aztreonam and cefepime, with MIC₅₀ values of 64 and 128 μ g/mL, respectively (Table S1). Of all isolates, 17 were resistant to colistin with MIC ≥ 4 μ g/mL.

All 66 isolates co-harboured *bla*_{OXA-23} and intrinsic *bla*_{OXA-51-like}. Among these, 34 isolates harboured *bla*_{OXA-66}, eight harboured *bla*_{OXA-104}, seven harboured *bla*_{OXA-64}, six harboured *bla*_{OXA-371}, four harboured *bla*_{OXA-69}, four harboured *bla*_{OXA-91}, two harboured *bla*_{OXA-68} and one harboured *bla*_{OXA-51}. Additionally, 26 isolates harboured *bla*_{NDM-1}, and 14 harboured *bla*_{PER-7}. Regarding aminoglycoside-resistant genes, 55 isolates harboured *armA*, 20 harboured *aadA1*, 16 harboured *aac(6')-Ib* and five harboured *aadB*. All isolates had point mutations in the quinolone-resistance-determining regions of *gyrA* and *parC*, consisting of the amino acid substitutions Ser83Leu and Asp87Glu in *GyrA*, and Ser80Leu in *ParC*.

The frequencies of isolates harbouring genes associated with biofilm formation were 100% for *cusE*, *bfmS*, *leuO*, and *ompA*, 94% for *pgaB*, and 92% for *epsA/ptk*.

3.2. MLST and phylogenetic analysis

The 66 MDR *A. baumannii* isolates belonged to eight different MLST ST (Table 1). eBURST analysis, combined with the Institute Pasteur MLST database, grouped four of the eight ST into three CCs, CC1 (ST1 and ST623), CC2 (ST2), and CC149 (ST149). CC1 contained 10 isolates (nine ST1 and one ST623), CC2 contained 34 isolates (all ST2) and CC149 contained eight isolates (all ST149). A maximum-likelihood phylogenetic tree constructed from the 66 MDR isolates was congruent with the tree constructed by the Bayesian method (Fig. 1). These phylogenetic trees revealed three clades corresponding to CC1, CC2, and CC149. As shown in Fig. 1, all 66 isolates encoded *bla*_{OXA-51-like} variants specific to their respective ST. Of the 10 isolates belonging to CC1, four ST1 isolates had *bla*_{OXA-69}, five ST1 isolates, and one ST623 isolate had *bla*_{OXA-371}. All 34 CC2 isolates (ST2) had *bla*_{OXA-66}, and all eight isolates belonging to ST149 had *bla*_{OXA-104}. The majority of CC2 and all CC149 isolates harboured *bla*_{NDM-1} (Table 1).

Table 1
Characterization of multidrug-resistant *Acinetobacter baumannii* isolates (n = 66).

MLST	No. of isolates	No. of strains harbouring the gene/total no. of strains				Mutations in QRDRs	
		Carbapenemase-encoding gene(s)	ESBL-encoding gene(s)	16S rRNA methyltransferase genes	Aminoglycoside acetyl/adenylyl transferase-encoding genes	GyrA	ParC
ST1	9	<i>bla</i> _{NDM-1} (3/9), <i>bla</i> _{OXA-23} <i>bla</i> _{OXA-69} (4/9), <i>bla</i> _{OXA-371} (5/9)	<i>bla</i> _{PER-7} (2/9)	<i>armA</i> (4/9)	<i>aadA1</i> (4/9), <i>aadB</i> (4/9)	S83L, D87E	S80L
ST2	34	<i>bla</i> _{NDM-1} (15/34), <i>bla</i> _{OXA-23} (33/34) <i>bla</i> _{OXA-66}	–	<i>armA</i> (32/34)	<i>aac-lb</i> (16/34), <i>aadA1</i> (16/34)	S83L, D87E	S80L
ST10	2	<i>bla</i> _{OXA-23} , <i>bla</i> _{OXA-68}	<i>bla</i> _{PER-7} (2/2)	<i>armA</i>	–	S83L, D87E	S80L
ST15	1	<i>bla</i> _{OXA-23} , <i>bla</i> _{OXA-51}	–	<i>armA</i>	–	S83L, D87E	S80L
ST25	7	<i>bla</i> _{OXA-23} , <i>bla</i> _{OXA-64}	<i>bla</i> _{PER-7} (7/7)	<i>armA</i>	–	S83L, D87E	S80L
ST149	8	<i>bla</i> _{NDM-1} , <i>bla</i> _{OXA-23} , <i>bla</i> _{OXA-104}	<i>bla</i> _{PER-7} (3/7)	<i>armA</i> (6/8),	–	S83L, D87E	S80L
ST164	4	<i>bla</i> _{OXA-23} , <i>bla</i> _{OXA-91}	–	<i>armA</i> (3/4)	<i>aadB</i> (1/4)	S83L, D87E	S80L
ST623 (CC1)	1	<i>bla</i> _{OXA-23} , <i>bla</i> _{OXA-371}	–	–	–	S83L, D87E	S80L

QRDRs: quinolone resistance-determining regions.

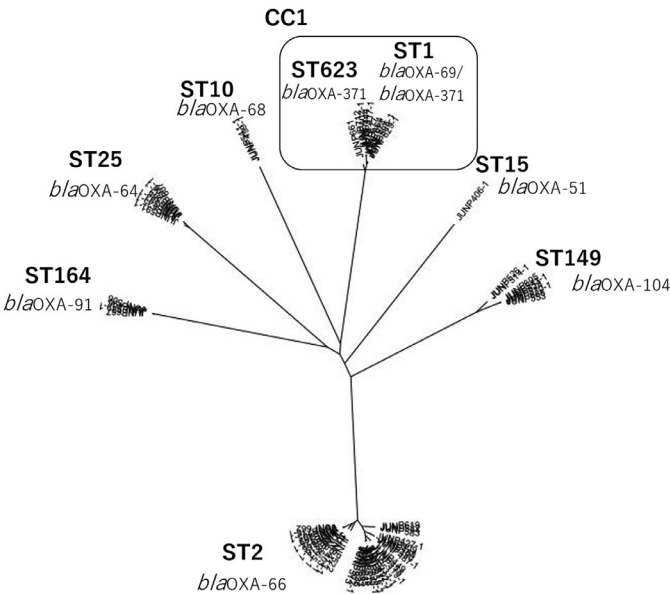


Fig. 1. Molecular phylogenetic tree of the 66 MDR *Acinetobacter baumannii* strains isolated in a medical setting in Nepal. The *bla*_{OXA-51}-like genes encoding intrinsic OXA-type carbapenemase differed depending on STs. Of all 66 isolates, the majority of 34 isolates belonged to ST2.

3.3. Genomic environments surrounding *bla*_{NDM-1} and *armA*

The genomic environments surrounding *bla*_{NDM-1} revealed three types of genetic structures (Fig. 2A–C). Type A structure was IS*Aba125*-*orf*-IS*S91*-*groL*-*groES*-*orf*-*tat*-*orf*-*ble*_{MBL}-*bla*_{NDM-1}-IS*Aba125*-*traD*-*orf*-*mobL*-*aph*(6)-*Id*-*aph*(3'')-*lb*-IS*Aba1* in ST1 and several ST2 isolates. This structure showed 100% identity with that of *A. baumannii* strain VB473 (GenBank accession no. CP050388) in India and *A. baumannii* strain AR_0083 chromosome (GenBank accession no. CP027528) isolated in the United States. Type B structure was IS*Aba125*-*orf*-IS*S91*-*groL*-*groES*-*cutA*-*dsbD*-*orf*-*ble*_{MBL}-*bla*_{NDM-1}-IS*Aba125*-*orf*-*orf*-*orf*, found in several ST2 and ST149 isolates, showing 100% identity *A. baumannii* strain KSK20 (GenBank accession no. CP072300) in India and *A. baumannii* strain IOMTU433 (GenBank accession no. AP014649) in Nepal. Type C structure was IS*Aba125*-*orf*-IS*S91*-*groL*-*cutA*-*orf*-*orf*-*bla*_{NDM-1}-IS*Aba125*-*orf*-*phS*-*pheT*-*ihfA*, found in several ST149 isolates, showing 100% identity with *A. baumannii* strain B11911

(GenBank accession no. CP021345) in India. Type A3 included JUNP543. Long-read sequencing revealed that *bla*_{NDM-1} was located on chromosomes in all five sequenced isolates (GenBank accession no. AP031576, AP031577, AP031578, AP031583 and AP031585).

The genomic environments surrounding *armA* revealed three types of genetic structures (Fig. 2D–F). Type D structure was IS*S26*-*repB*-*orf*-*mph*(E)-*msr*(E)-IS*Ec29*-*orf*-*orf*-*armA*-*orf*-IS*S10A*-IS*S5*-IS*S91*, found in ST1, ST10, ST15, ST25, and ST149 isolates. This structure was identical to those in *A. baumannii* strain AR_0083 chromosome (GenBank accession no. CP027528) in the United States and *A. baumannii* strain AB181-VUB chromosome, (GenBank accession no. CP091358) in Belgium. Type E structure was IS*Aba24*-*orf*-*mph*(E)-*msr*(E)-IS*Ec29*-*orf*-*orf*-*armA*-*orf*-IS*S5*-IS*S91*, found in several ST2 isolates, showing 100% identity with *A. baumannii* strain DT-Ab003 (GenBank accession no. CP050916) in China and *Salmonella enterica* subsp. *enterica* serovar Stanley strain AR-0403 chromosome, (GenBank accession no. CP044184) in the United States. Type F structure was IS*Aba24*-*orf*-*mph*(E)-*msr*(E)-IS*Ec29*-*orf*-*orf*-*armA*-*orf*-IS*S5*-*pqiC*-*orf*, found in several ST2 and ST164 isolates, showing 100% identity with *A. baumannii* strain VB473 (GenBank accession no. CP050388) in India. Long-read sequencing revealed that *armA* was located on chromosomes in ST1 (GenBank accession no. AP031576 and AP031588), ST2 (GenBank accession no. AP031577 and AP031578), ST10 (GenBank accession no. AP031582), ST15 (GenBank accession no. AP031579), and ST164 isolates (GenBank accession no. AP031581), but on 184–191 kbp plasmids in ST25 (GenBank accession no. AP031587) and ST149 isolates (GenBank accession no. AP031584).

4. Discussion

The findings of this study indicate that *A. baumannii* strains highly resistant to carbapenems and/or aminoglycosides are spreading in medical settings in Nepal. The genetic backgrounds of *A. baumannii* clinical isolates in Nepal appear to have markedly changed between 2012 and 2020. In our previous study conducted in 2012 in Nepal, the frequencies of CC1, CC2, and CC149 *A. baumannii* were 17%, 30%, and 36%, respectively. However, in the present study from 2020, the frequencies of CC1, CC2, and CC149 changed to 15%, 52%, and 14%, respectively [11]. Specifically, the frequencies of ST623 belonging to CC1 and ST149 isolates decreased significantly from 14% to 2% and from 31% to 14%, respectively. These results suggest that the CC1 (international clone 1)

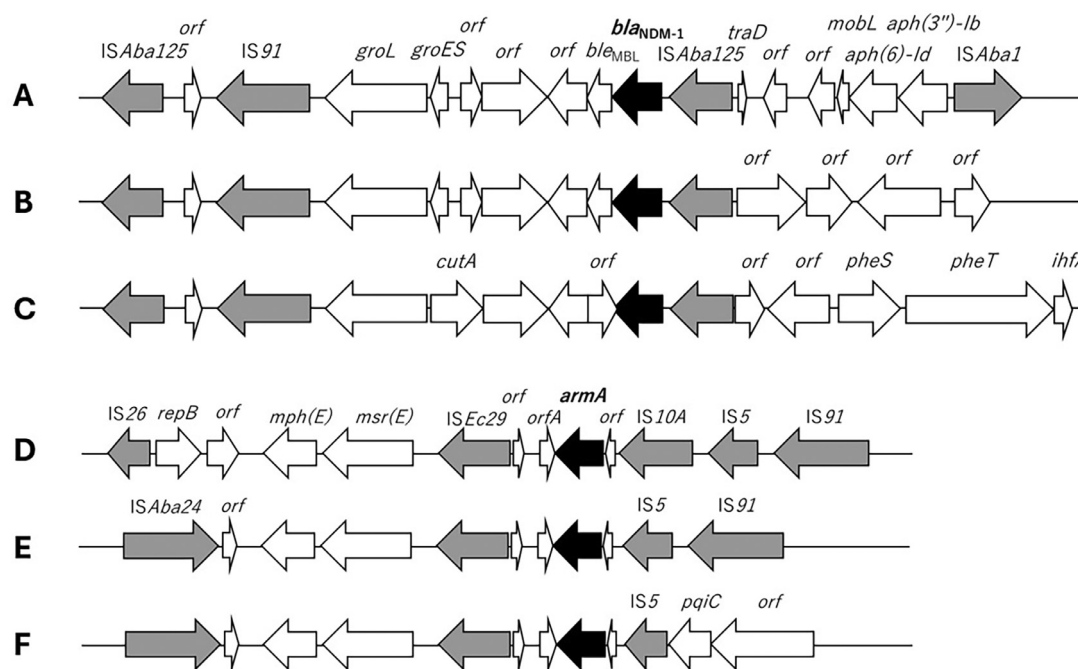


Fig. 2. Genomic environments surrounding *bla*_{NDM-1} and *armA* in MDR *Acinetobacter baumannii* in Nepal. The structures of *bla*_{NDM-1} had 3 types (A–C) and *armA* had 3 types (D–F). The mobile genetic elements are shaded in grey and genes encoding NDM-1 and 16S rRNA methyltransferase, *ArmA*, are shaded in black.

and CC149 clones have been replaced by CC2 (international clone 2) over several years.

The frequency of MDR *A. baumannii* clinical isolates harbouring *bla*_{NDM-1} and *armA* has increased from 2012 to 2020 in Nepal [11]. In 2012, the frequencies of isolates harbouring *bla*_{NDM-1} and *armA* were 26% and 58%, respectively, whereas in 2020, they had increased to 39% and 83%, respectively [11]. There are few reports on antibiotic use in Nepalese healthcare facilities. One study reported that the prevalence of antibiotic use among hospitalized COVID-19 patients was higher compared to other studies conducted in similar settings [12]. The use of antibiotics in medical settings may be a contributing factor to the increased isolation rate of pathogens harbouring carbapenem- and aminoglycoside-resistant genes.

A. baumannii with biofilm formation and drug resistance is likely spreading in medical settings in Nepal. Biofilm formation in *Acinetobacter* spp. is associated with adherence to human cells (e.g. epithelial cells and/or pneumocytes) and cell invasion [13]. Recent studies have reported that LeuO, a LysR-type transcriptional regulator, is involved in biofilm formation and virulence of *A. baumannii* [14]. Our study suggests that *A. baumannii* clinical isolates with multidrug resistance and virulence are disseminating in Nepal due to the high prevalence of genes associated with multidrug resistance, biofilm formation, and virulence.

Acknowledgements

This study was approved by the Ethical Committee, Institute of Medicine, Tribhuvan University (approval number: 219 (6-11-E)2/075/076), and the Biosafety Committee, Juntendo University (approval number: BSL2/29-1). We thank David Price of English Services for Scientists for proofreading.

Funding: This work was supported by Japan Society for the Promotion of Science (grant numbers: 21K07031, 23K06550 and 24K10204).

Ethical approval: Not required.

Declaration of competing interests: None declared.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2024.07.017](https://doi.org/10.1016/j.jgar.2024.07.017).

References

- [1] Peleg AY, Seifert H, Paterson DL. *Acinetobacter baumannii*: emergence of a successful pathogen. Clin Microbiol Rev 2008;21(3):538–82. doi:10.1128/CMR.00058-07.
- [2] Bergogne-Berezin E, Towner KJ. *Acinetobacter* spp. as nosocomial pathogens: microbiological, clinical, and epidemiological features. Clin Microbiol Rev 1996;9(2):148–65. doi:10.1128/CMR.9.2.148.
- [3] Moubareck CA, Halat DH. Insights into *Acinetobacter baumannii*: a review of microbiological, virulence, and resistance traits in a threatening nosocomial Pathogen. Antibiotics (Basel) 2020;9(3):119. doi:10.3390/antibiotics9030119.
- [4] Hamouda A, Amyes SG. Novel gyrA and parC point mutations in two strains of *Acinetobacter baumannii* resistant to ciprofloxacin. J Antimicrob Chemother 2004;53(4):695–6. doi:10.1093/jac/dkh368.
- [5] Wong D, Nielsen TB, Bonomo RA, Pantapalangkoor P, Luna B, Shellberg B. Clinical and pathophysiological overview of *Acinetobacter* infections: a century of challenges. Clin Microbiol Rev 2017;30(1):409–47. doi:10.1128/CMR.00058-16.
- [6] Lupo A, Haenni M, Madec JY. Antimicrobial resistance in *Acinetobacter* spp. and *Pseudomonas* spp. Microbiol Spectr 2018;6(3). doi:10.1128/microbiolspec.ARBA-0007-2017.
- [7] Kyriakidis I, Vasileiou E, Pana ZD, Tragiannidis A. *Acinetobacter baumannii* antibiotic resistance mechanisms. Pathogens 2021;10(3):373. doi:10.3390/pathogens10030373.
- [8] Performance Standards for Antimicrobial Susceptibility Testing. CLSI supplement M100 clinical and laboratory standards institute. 33rd Edition. Wayne, PA: Clinical and Laboratory Standards Institute; 2023.
- [9] Piewngam P, Kiratisin P. Comparative assessment of antimicrobial susceptibility testing for tigecycline and colistin against *Acinetobacter baumannii* clinical isolates, including multidrug-resistant isolates. Int J Antimicrob Agents 2014;44(5):396–401.
- [10] Wick RR, Judd LM, Gorrie CL, Holt KE. Unicycler: resolving bacterial genome assemblies from short and long sequencing reads. PLoS Comput Biol 2017;13(6):e1005595. doi:10.1371/journal.pcbi.1005595.
- [11] Shrestha S, Miyoshi-Akiyama T, Ohara H, Shimada K, Satou K, Teruya K, et al. Molecular epidemiology of multidrug-resistant *Acinetobacter baumannii* isolates in a university hospital in Nepal reveals the emergence of a novel epidemic clonal lineage. Int J Antimicrob Agents 2015;46(5):526–31. doi:10.1016/j.ijantimicag.2015.07.012.

- [12] Thapa B, Pathak SB, Jha N, Sijapati MJ, Shankar PR. Antibiotics use in hospitalised COVID-19 patients in a tertiary care centre: a descriptive cross-sectional study. *JNMA J Nepal Med Assoc* 2022;60(251):625–30. doi:[10.31729/jnma.7394](https://doi.org/10.31729/jnma.7394).
- [13] Lucidi M, Visaggio D, Migliaccio A, Cappecchi G, Visca P, Imperi F, et al. Pathogenicity and virulence of *Acinetobacter baumannii*: factors contributing to the fitness in healthcare settings and the infected host. *Virulence* 2024;15(1):2289769. doi:[10.1080/21505594.2023.2289769](https://doi.org/10.1080/21505594.2023.2289769).
- [14] Islam MM, Kim K, Lee JC, Shin M. LeuO, a LysR-type transcriptional regulator, is involved in biofilm formation and virulence of *Acinetobacter baumannii*. *Front Cell Infect Microbiol* 2021;11:738706. doi:[10.3389/fcimb.2021.738706](https://doi.org/10.3389/fcimb.2021.738706). eCollection 2021.